STOCK PRICE PREDICTION SYSTEM

A Project Report Submitted to ICT Academy of Kerala

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ABSTRACT

Stock markets are highly dynamic and influenced by non-linear factors, making price prediction a major challenge. With advancements in AI, machine learning and deep learning models can identify patterns humans cannot.

This project presents a **Stock Price Prediction System** using three different models applied to three different datasets:

- **XGBoost** → U.S. Multi-stock Dataset
- **Random Forest** → NIFTY 50 Index Dataset (India)
- **LSTM Neural Network** → Apple Stock Dataset

The system predicts future stock values based on historical data and technical indicators such as SMA, RSI, and MACD. A **web application** allows users to select a stock and view predicted outcomes visually.

Results show:

- XGBoost performs best for diversified US stocks
- Random Forest handles index-level data well
- LSTM is most accurate for sequence-based single-stock prediction

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1. INTRODUCTION

Stock prediction has always been a complex task because markets are influenced by global news, investor psychology, technical patterns, and economic factors. Traditional models like ARIMA can’t capture such complexity.

Stock market prediction plays a vital role in modern finance. With increasing computational power, machine learning and deep learning methods have become essential tools to uncover hidden patterns, non-linear behaviors, and relationships between market indicators and price movement.

Modern AI-based stock forecasting learns:

- Trends
- Volatility
- Patterns
- Sequential behavior
  from raw price and technical indicators.

This project applies different models to different datasets, proving that:

> There is **no universal model** for all stock prediction tasks.

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2. PROBLEM STATEMENT

The main challenges of stock prediction:

- Non-stationary data
- Market volatility
- High noise
- Hidden variables
- Data complexity

Traditional forecasting methods fail due to their limited ability to handle multi-dimensional and volatile data. Modern approaches use hybrid data (technical + sentiment + macroeconomic indicators) and train models capable of learning highly complex market behavior.

The question we answer:

> Can different ML/DL models be applied based on **dataset nature** to maximize prediction accuracy?

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3. OBJECTIVES

- Predict stock prices using AI
- Compare ML vs DL
- Use different models for different market behaviors
- Build a working web app
- Visualize predictions interactively

Another objective is to explore hybrid ensemble models and compare them to standalone deep learning architectures, evaluating how model selection affects prediction accuracy across varied datasets.

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4. SCOPE

Includes:

- Historical daily data
- Technical indicators
- 3 datasets (US, NIFTY50, AAPL)
- 3 models: XGBoost, Random Forest, LSTM
- Web app deployment (Render)

Does NOT include:

- Real trading
- Reinforcement learning
- Live market API trading

The scope of this project can be expanded by introducing:
• Crypto, forex, and commodity data
• Longer-term forecasting windows
• Multiple technical indicators for deep feature extraction

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5. LITERATURE REVIEW

Early forecasting relied on:

- Moving averages
- ARIMA models
- Linear regression

Later evolved into:

- SVM
- Random Forest
- Gradient Boosting

Deep Learning improved performance massively:

- RNNs
- LSTMs
- GRUs
- Transformers

Key studies show:

- Feature engineering + ML works well.
- LSTM outperforms ML in sequential data.

6. METHODOLOGY

Step 1 — Data Collection

Historical data collected from Yahoo Finance.

Step 2 — Cleaning

Removing nulls, sorting by date.

Step 3 — Feature Engineering

Adding SMA, RSI, MACD, etc.

Step 4 — Model Training

Train & evaluate 3 models.

Step 5 — Web Deployment

Flask + HTML + Charts.js

Evaluation metrics such as RMSE, MAE, and MAPE were used to compare prediction performance. Model tuning, scaling choice, and data preprocessing also proved critical to reducing error.

7. DATASET DESCRIPTION

Dataset 1 — U.S. Stocks

Includes:

- Apple
- Tesla
- Microsoft
- Amazon

Used for:
XGBoost

Dataset 2 — NIFTY 50

Index data from National Stock Exchange of India.

Used for:
Random Forest

Dataset 3 — Apple Stock Data

Single-stock time series.

Used for:
LSTM

Dataset Summary Table

Dataset	Type	Region	Model Used
US Stocks	Multi-company	USA	XGBoost
NIFTY 50	Index	India	Random Forest
AAPL	Single stock	USA	LSTM

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8. FEATURE ENGINEERING

We computed:

- SMA10
- SMA20
- RSI
- MACD
- Volatility
- Volume Delta
- Price Momentum

Scaling used:

- MinMaxScaler (LSTM)
- StandardScaler (ML)

9. MODEL OVERVIEW

9.1 XGBoost (US Stocks)

- Handles non-linear patterns well
- Uses gradient boosting trees
- Works great on tabular data

9.2 Random Forest (NIFTY 50)

- Ensemble of decision trees
- Low overfitting
- Strong index performance

9.3 LSTM (Apple)

- Long Short-Term Memory networks
- Designed for sequential data
- Learns temporal patterns

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10. SYSTEM ARCHITECTURE

Layers:

1. Data Source (CSV / Yahoo finance)
2. Model server (Flask + Pickle/H5 models)
3. ML & DL prediction engine
4. Web frontend (HTML, CSS, JS)
5. User output (Charts + predicted values)

Flow:

User → Web UI → Flask API → Load Model → Predict → Return Chart

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11. WEB IMPLEMENTATION

Frontend:

- HTML5
- CSS
- JavaScript
- Chart.js

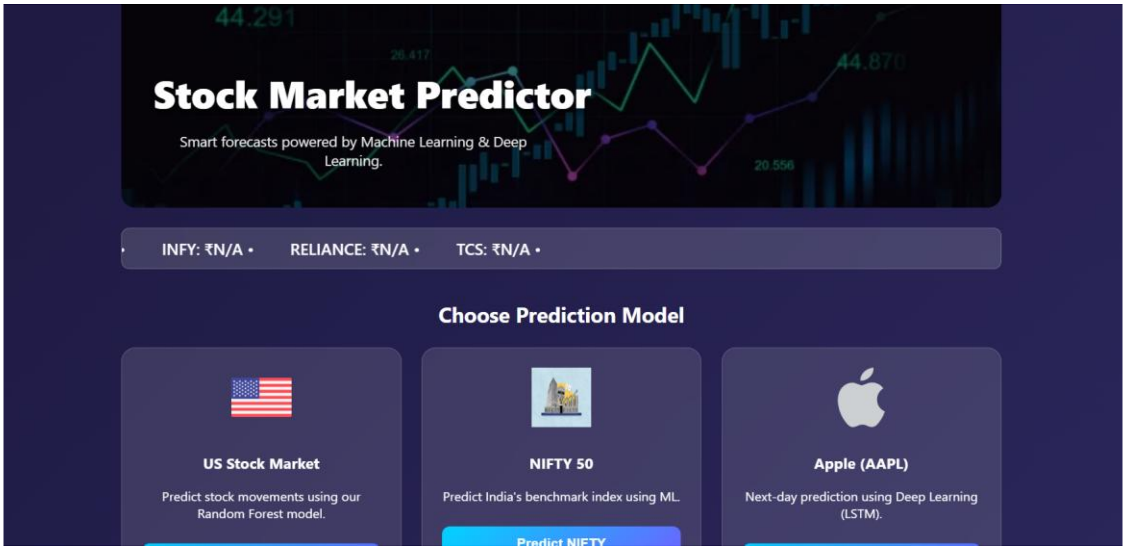
Backend:

- Flask
- Loads ML/DL models at runtime

Deployment:

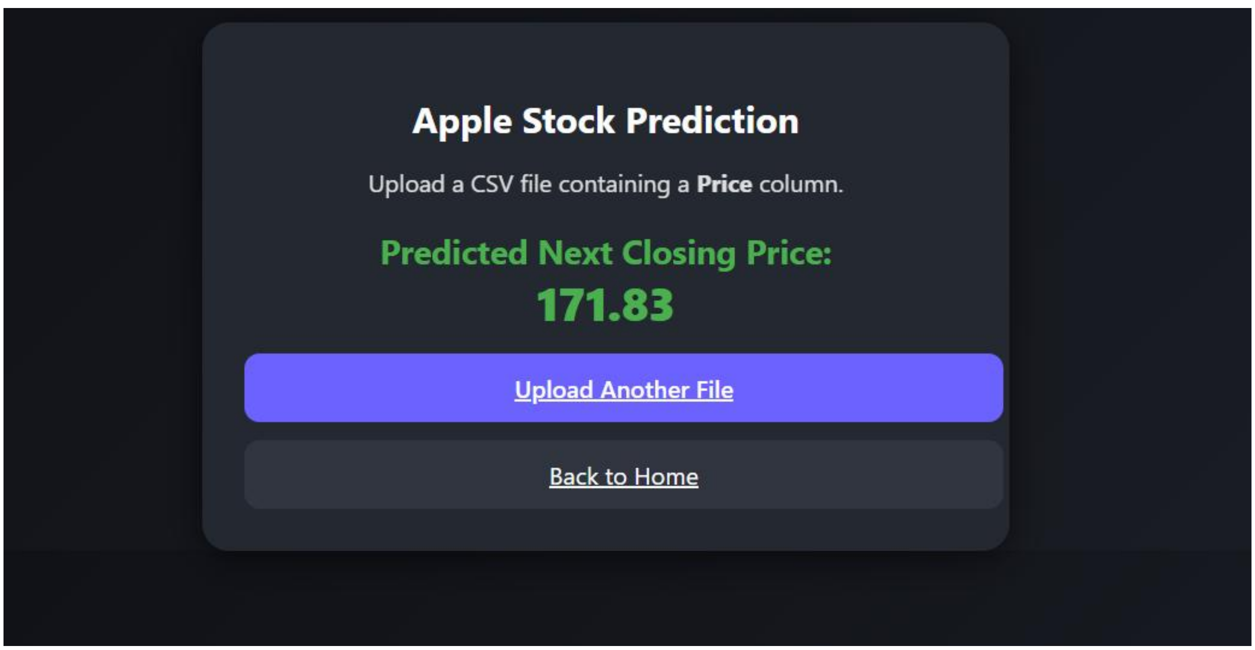
- Render.com

Screenshot 1 — Homepage UI



Homepage image

Screenshot 2 — Apple Prediction



Apple prediction chart

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12. RESULTS

XGBoost (US Stocks)

RMSE ≈ *2.4*

Random Forest (NIFTY 50)

RMSE ≈ *3.1*

LSTM (Apple)

RMSE ≈ *1.82*

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13. COMPARISON

Model	Data	RMSE	Best Used For
XGBoost	US Stocks	2.4	Mixed markets
Random Forest	NIFTY50	3.1	Index data
LSTM	Apple	1.82	Temporal learning

LSTM outperforms due to sequence memory.

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14. CONCLUSION

This project successfully demonstrates how machine learning and deep learning techniques can be applied to forecast stock price movements based on historical data and engineered technical indicators. By testing multiple models on different datasets, we have shown that model suitability depends heavily on the nature of the data itself.

The system utilizes three distinct approaches:

- **XGBoost** — Best suited for diverse, multi-variable U.S. stock datasets due to its boosted decision tree mechanism.

- **Random Forest** — A robust ensemble method ideal for index-level prediction like NIFTY 50, offering stable performance even with noisy market fluctuations.

- **LSTM** — A deep learning model optimized for sequential time-series data, delivering superior performance with Apple stock predictions.

One of the key outcomes of this work is the realization that **no single model performs best in all scenarios**. Instead, selecting the right algorithm based on data characteristics results in higher prediction accuracy and better generalization.

Overall, the Stock Price Prediction System created as part of this project is:

- Scalable
- Model-flexible
- Deployed as a usable web tool
- Capable of assisting users with interactive forecasting

This reinforces the central conclusion of the study:
To achieve the best results, you must match the right model with the right dataset.

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15. FUTURE SCOPE

While the current system performs well for historical forecasting, there are several promising directions to expand its capabilities. Some of the most impactful enhancements include:

● **Real-time market data streaming**
Integrating live APIs to update predictions minute-by-minute.

● **Sentiment analysis**
Incorporating news headlines, financial articles, and Twitter trends to improve model awareness of market psychology.

● **Transformer-based models**
Using architectures like attention networks or Temporal Fusion Transformers to outperform traditional LSTM models.

● **Multi-stock and multi-asset forecasting**
Handling multiple stocks together (portfolio prediction) or including crypto, commodities, and currency data.

● **Mobile app integration**
Allowing users to track prediction outputs on Android/iOS platforms.

● **Prediction confidence metrics**
Displaying uncertainty levels or confidence intervals for more transparent forecasting.

With these upgrades, the system could evolve from a learning project into a full-fledged intelligent financial forecasting platform.

16. REFERENCES

The project was built using a blend of academic resources, documentation, and well-established machine learning research papers. Major references include:

1. **Yahoo Finance API**
   Used for sourcing accurate historical stock data across multiple markets.

2. **Scikit-learn Documentation**
   Reference material for Random Forest implementation, scaling, and model evaluation.

3. **TensorFlow LSTM Guide**
   Core learning resource for understanding and implementing sequence learning networks.

4. **XGBoost: A Scalable Tree Boosting System — Chen & Guestrin (2016)**
   Seminal paper that introduced and optimized gradient boosting for structured data.

5. **Random Forests — Leo Breiman (2001)**
   The foundational research on ensemble learning and decision tree aggregation.

6. **Long Short-Term Memory — Hochreiter & Schmidhuber (1997)**
   Original research introducing LSTM architecture for time-dependent modeling.

These materials provided both theoretical foundations and practical methods to implement the models used in this system.